

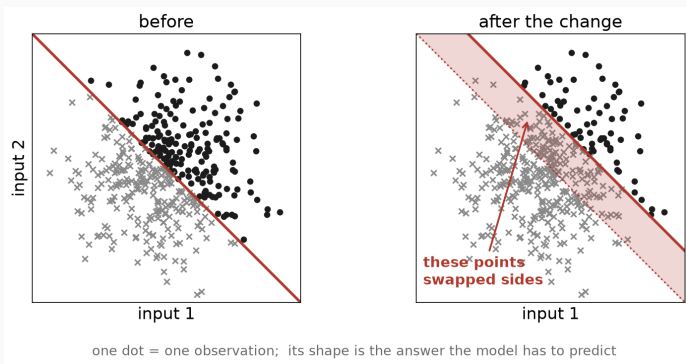
The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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Concept drift: the rule changes, the data does not



A model trained on the left keeps predicting, confidently, and is now **wrong on the shaded band**.

Defence one: a model that repairs itself

- A **decision tree** splits the picture with simple cuts, and answers by asking which piece a new point falls into.
- One tree is fragile, so ten of them **vote**. That is the Adaptive Random Forest, the model we study.
- **It repairs itself**: a tree whose answers start going wrong is thrown away, and a fresh one grows in its place.

Ten trees, replaced one at a time. Remember that number.

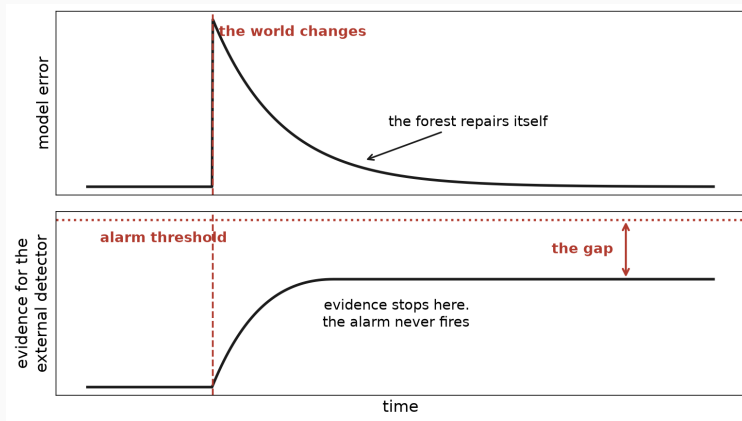
Defence two: a watchdog that raises the alarm

- It reads the model's mistakes, one observation at a time, and adds up the excess.
- Once the pile crosses a threshold, it calls a human: audit, retraining, pause.

The alarm is the only thing a human ever sees.

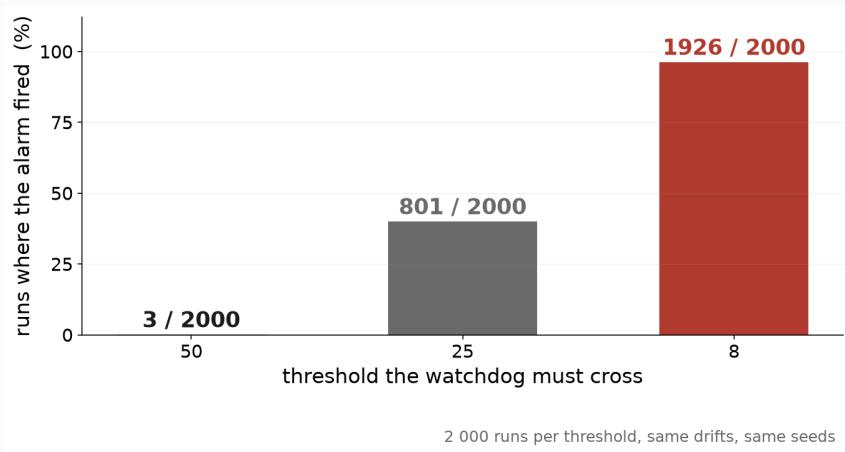
Two communities, two tools, never measured against each other.

Put them together, and they cancel out



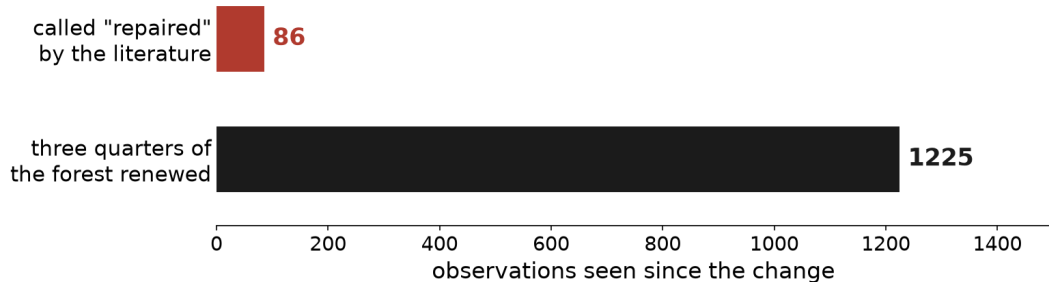
schematic

Why not just lower the threshold?



A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

Our contribution: the accepted clock is wrong



median of 100 runs, forest of 10 trees, change size 0.24

It marks the first tree out of ten, not a repaired forest.

Where we stand

- The paradox is **real and reproducible**, and we can say when the evidence falls short of what the alarm needs.
- The clock the field relies on is **biased, always the same way**. Every conclusion drawn from it inherits that bias.
- **Next:** watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.